DAPI Registration for cycIF Fold QC

Mechanics of cross-round alignment, and how to wire it into a between-round fold-detection pipeline

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Purpose of this document

This is a reference for the **registration step** of your fold-QC idea: detecting tissue that folds, lifts, or warps *between* staining rounds by comparing each round's DAPI image against a reference. Registration is the enabling step — it removes the nuisance motion (the slide shifting on the stage) so that the residual disagreement you measure afterward is real tissue change and not stage wobble. The one counter-intuitive point to carry through everything below: for *this* purpose you deliberately want a **weak (rigid) registration**, not a strong one.

1. What you are actually correcting

Between rounds, the slide is removed from and returned to the stage (or the stage re-homes). The DAPI image of a given core therefore differs from the previous round for a few physically distinct reasons. Separating them is the whole design:

- **Translation** — the core sits at a slightly different (x, y). Almost always the largest term.
- **Small rotation** — the slide re-seats at a slightly different angle.
- **Slight scale** — focus / magnification drift. Usually negligible.
- **Local non-rigid deformation** — the fold, lift, or warp itself. **This is your signal.**

The global part (translation + rotation) is **nuisance** you want to remove. The local non-rigid part is **signal** you want to keep and measure. Registration must be strong enough to kill the nuisance but not so strong that it silently absorbs the signal. That tension drives every choice that follows.

2. The transform hierarchy

Registration transforms form a ladder of increasing flexibility. The rule: choose the **least** flexible transform that still removes your nuisance motion.

Transform	Free params	Can represent	Right for you?
Translation	2 (dx, dy)	Shift only	Global step (if no rotation)
Rigid	3 (dx, dy, θ)	Shift + rotation, distances preserved	**YES — correct model**
Affine	6	+ scale + shear	No — too permissive
Non-rigid	thousands	Every region moves freely; a fold	**NO — erases signal**

Rigid is the physically honest model for cycIF stage motion: the tissue did not scale or shear on the stage, it moved and rotated. Reaching for affine “to be safe” gives the model freedom to explain real deformation away as global shear — imprecision worth resisting.

3. How alignment is computed — two families

3a. Frequency-domain (phase correlation)

For pure translation. A shift between two images becomes a linear phase ramp in Fourier space; you recover it as a single sharp peak in the inverse-transformed cross-power spectrum. Fast, robust, sub-pixel. This is your cheap, reliable estimator of the global shift. Log-polar variants also recover rotation and scale.

```
from skimage.registration import phase_cross_correlation
from scipy.ndimage import shift as nd_shift

# ref, mov: 2D DAPI arrays (reference round, moving round), same core
shift_yx, error, phasediff = phase_cross_correlation(
    ref, mov, upsample_factor=10)      # sub-pixel: 1/10 px
mov_registered = nd_shift(mov, shift=shift_yx, order=1)
print("estimated (dy, dx) shift:", shift_yx)
```

3b. Intensity-based iterative optimization

The general workhorse. You specify four components and an optimizer searches transform parameters that maximize image similarity:

- **Transform model** — rigid / affine / B-spline (pick rigid, see §2).
- **Similarity metric** — what "aligned" means to the optimizer (see §4).
- **Optimizer** — gradient descent etc., nudging params iteratively.
- **Multi-resolution pyramid** — solve on a downsampled image first (fast, avoids local minima), then refine at finer resolution.

```
import SimpleITK as sitk

fixed  = sitk.GetImageFromArray(ref.astype('float32'))
moving = sitk.GetImageFromArray(mov.astype('float32'))

R = sitk.ImageRegistrationMethod()
R.SetMetricAsCorrelation()                       # NCC: same-modality DAPI
R.SetOptimizerAsRegularStepGradientDescent(
        learningRate=1.0, minStep=1e-4, numberOfIterations=200)
R.SetInitialTransform(sitk.Euler2DTransform())  # RIGID: dx, dy, theta
R.SetInterpolator(sitk.sitkLinear)
R.SetShrinkFactorsPerLevel([4, 2, 1])           # pyramid
R.SetSmoothingSigmasPerLevel([2, 1, 0])

tx = R.Execute(fixed, moving)
resampled = sitk.Resample(moving, fixed, tx, sitk.sitkLinear, 0.0)
mov_registered = sitk.GetArrayFromImage(resampled)
```

4. The similarity metric — one precision point

You are aligning the **same stain (DAPI) in both rounds** — same modality — so intensities are directly comparable and **normalized cross-correlation (NCC)** is the natural metric. Know *why*: if you ever register DAPI to a different modality (H&E, brightfield), NCC breaks, because it assumes an approximately linear intensity relationship. There you would switch to **mutual information**, which only assumes a statistical relationship, not a linear one. Same-modality now → NCC; recognize the day that assumption stops holding.

5. Existing tooling

- **scikit-image** — `phase_cross_correlation`. Your lightweight global-shift step.
- **SimpleITK / SimpleElastix** — full explicit control of transform / metric / optimizer / pyramid. Best for understanding, not a black box.
- **ASHLAR** — cycIF-native (Sorger lab / MCMICRO), uses DAPI as the registration anchor by design. If you are near that pipeline it may already do your global step.
- **VALIS** — whole-slide multiplex, automates rigid → non-rigid. Powerful, but see the warning — its non-rigid stage is exactly what you must avoid for fold QC.

6. The trap — read this twice

If registration is non-rigid (deformable), it will warp the fold region to match the reference — and in doing so erase the very signal you are trying to detect. A deformable transform is powerful enough to explain a fold as “local deformation” and align it away. Your disagreement map then comes back clean over a real fold: a false negative manufactured by an over-capable registration.

So for fold detection specifically, use **rigid (or translation-only)** global registration: strong enough to remove stage motion, too rigid to absorb a fold. The fold then **survives as residual** — exactly what you want to measure.

Note the inversion. Standard cycIF registration *does* use non-rigid warping, because its goal is to make channels line up for downstream single-cell analysis. Your QC goal is the opposite: preserve deformation so you can detect it. The “better” registration is *worse* for your purpose. If a colleague offers their non-rigid-registered images “to save you time,” their warp has already flattened the folds you are hunting — you must re-register from raw DAPI with a rigid model.

7. How this fits your pipeline

The registration step sits inside the larger two-reference fold-QC design you built:

- **Step 1 — Register.** Per core, align each round's DAPI to a reference with a **rigid** model. Start with `phase_cross_correlation` for the shift; step up to SimpleITK Euler2D if rotation is non-trivial. Removes stage motion.
- **Step 2 — Inspect the residual visually first.** Overlay reference and registered DAPI in two colors (R1 → magenta, R_n → green). Well-registered tissue goes grey/white; anything still colored is residual — folds, lifts, warps. Eyeball a few cores before trusting any scalar.
- **Step 3 — Per-tile residual disagreement.** Tile the registered images and compute a local disagreement metric (NCC or SSIM per tile). Do this both **sequentially** (R_n vs R_n–1, localizes *when* a fold appeared) and **vs. anchor** (R_n vs R1, catches slow cumulative drift).

- **Step 4 — Flag by the triple signature.** A real between-round fold is **local in space** (outlier vs. the rest of its own core), **abrupt in time** (a step above the rolling trend, not smooth wear), and **positive in sign** (local DAPI *increase* = doubled tissue). Diffuse mild *decrease* across a whole core is bleaching/degradation, not a fold.

> **Why register before comparing (Step 1 before Step 3):** without registration, global stage motion makes the whole image disagree, drowning the local fold signal in false positives. Register first, then the residual is real tissue change.

Appendix — residual overlay snippet

```
import numpy as np

def norm(x):
    x = x.astype('float32')
    lo, hi = np.percentile(x, [1, 99])
    return np.clip((x - lo) / (hi - lo + 1e-9), 0, 1)

# magenta = reference round, green = registered moving round
rgb = np.zeros((*ref.shape, 3), dtype='float32')
r, m = norm(ref), norm(mov_registered)
rgb[..., 0] = r          # R
rgb[..., 2] = r          # B  -> magenta = reference
rgb[..., 1] = m          # G  -> green    = moving
# grey/white = aligned; residual magenta or green = tissue change / fold
```

One open check to reason through before you build: your rigid registration itself can fail on a badly folded core (there may be no single rotation+shift that aligns it). How would you tell a registration failure apart from a detected fold — given both produce high residual disagreement?